AWS Observability and Monitoring

Objective

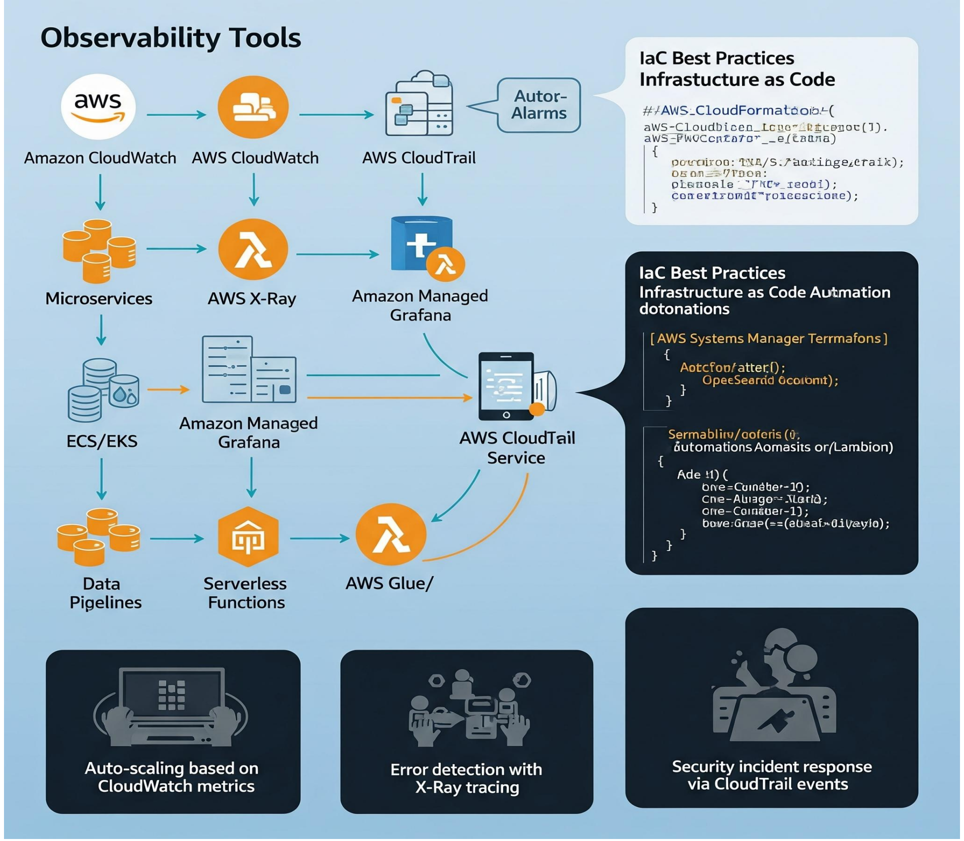
- ❖ Understand AWS observability tools
- ❖ Apply automation across workloads
- ❖ Explore real-world scenarios
- ❖ Learn IaC best practices
- ❖ Enable proactive operations

Why Observability & Automation Matter

- ❖ Manual monitoring is error-prone & slow
- ❖ Cloud introduces rapid scaling & complexity
- ❖ Automation removes repetitive tasks
- ❖ Observability ensures proactive detection
- ❖ Together they reduce downtime & cost

Key AWS Services

- ❖ Amazon CloudWatch
- ❖ AWS CloudFormation
- ❖ AWS Systems Manager
- ❖ Amazon EventBridge
- ❖ AWS Config
- ❖ AWS OpsWorks



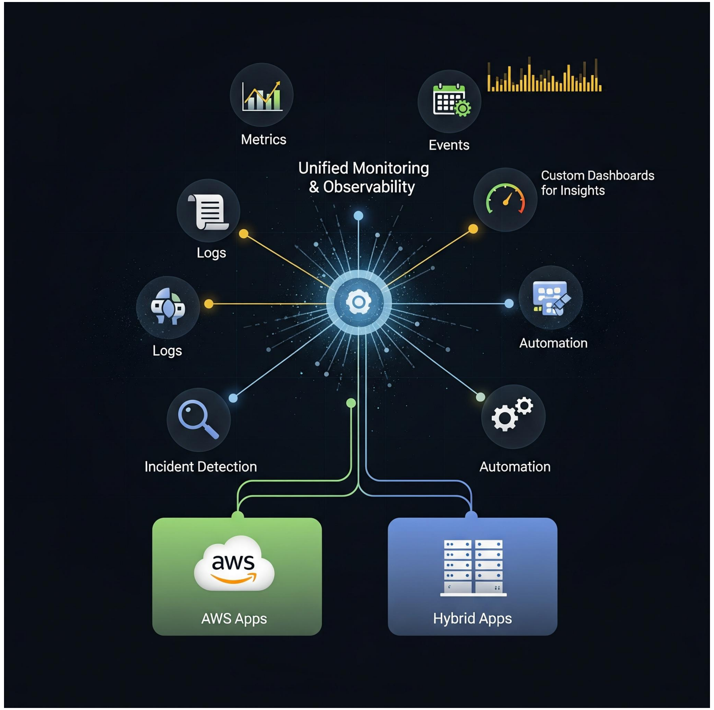
Infrastructure as Code (IaC) Context

- Infra defined in **templates/code**
- Enables **version control & repeatability**
- Automates provisioning at scale
- Reduces config drift & manual errors
- Key tools: CloudFormation, CDK, Terraform

AWS CloudWatch

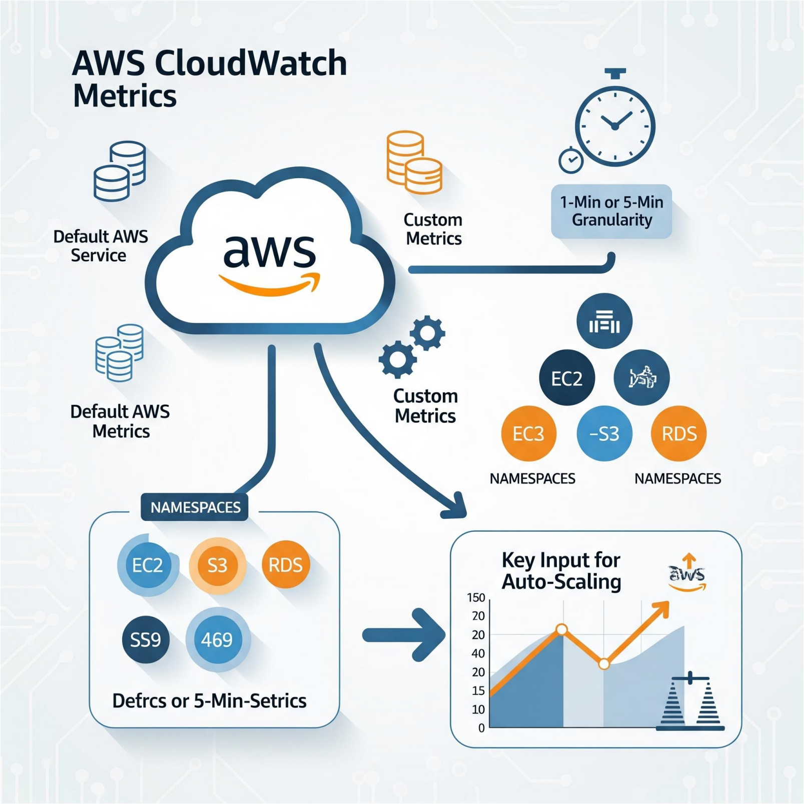
CloudWatch Overview

- ❖ Unified monitoring & observability
- ❖ Collects metrics, logs, events
- ❖ Custom dashboards for insights
- ❖ Works across AWS & hybrid apps
- ❖ Core for incident detection & automation



CloudWatch Metrics

- ❖ Default AWS service metrics included
- ❖ Custom metrics can be published
- ❖ Granularity: 1-min (detailed) or 5-min
- ❖ Metrics grouped in namespaces
- ❖ Key input for auto-scaling

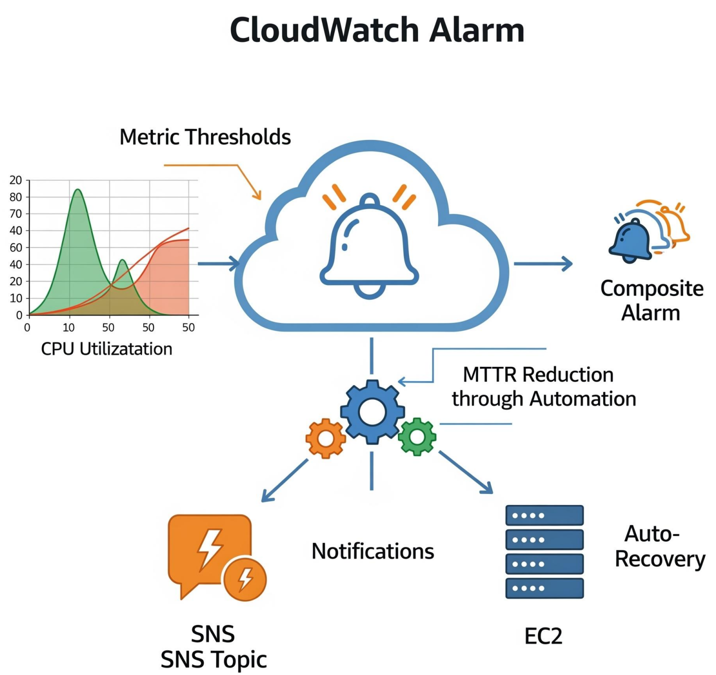


CloudWatch Logs

- Central log collection service
- Configurable retention policies
- Supports subscriptions (e.g., to Lambda)
- Search & analyze logs with Insights
- Send logs to S3 or third-party tools

CloudWatch Alarms

- ❖ Trigger on metric thresholds
- ❖ Integrates with SNS for notifications
- ❖ Can auto-recover EC2 or scale resources
- ❖ Composite alarms for multiple signals
- ❖ Reduces MTTR through automation

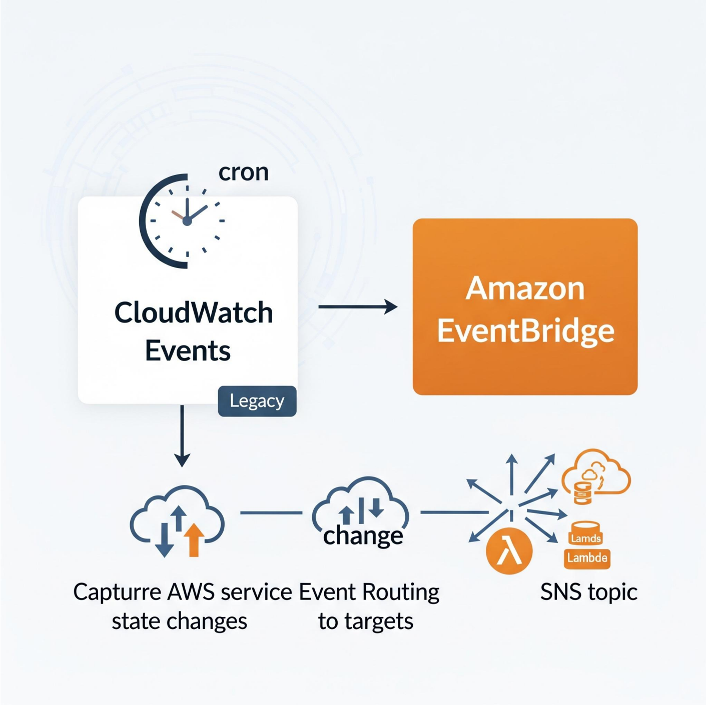


CloudWatch Dashboards

- ❖ Customizable visualization of metrics
- ❖ Multi-region & multi-account support
- ❖ Share dashboards with teams
- ❖ Real-time health monitoring
- ❖ Useful for NOC or SRE teams

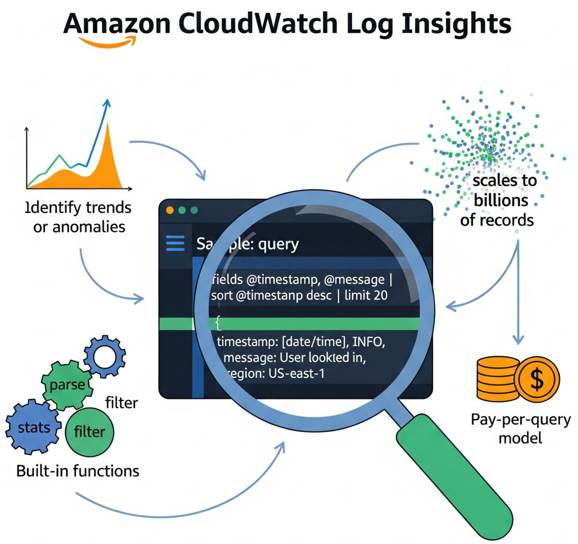
CloudWatch Events

- Legacy event bus (replaced by EventBridge)
- Supports cron-based rules
- Captures AWS service state changes
- Routes events to targets (Lambda, SNS)
- Key for event-driven automation



CloudWatch Log Insights

- ❖ Query language for log data
- ❖ Identify trends or anomalies
- ❖ Built-in functions (parse, stats, filter)
- ❖ Pay-per-query model
- ❖ Scales to billions of records



CloudWatch Synthetics

- Canaries simulate end-user behavior
- Detect issues before customers see them
- Scripts run on schedule or event
- Monitor APIs, endpoints, UI flows
- Integrated with alarms & dashboards

CloudWatch ServiceLens

- Correlates metrics, logs, and traces
- Integrates with AWS X-Ray
- Provides service maps for microservices
- Detects bottlenecks automatically
- Ideal for distributed architectures

CloudWatch Summary

- ❖ Core monitoring service in AWS
- ❖ Combines metrics, logs, events
- ❖ Enables auto-remediation workflows
- ❖ Integrates with scaling & security tools
- ❖ Foundation for observability strategy

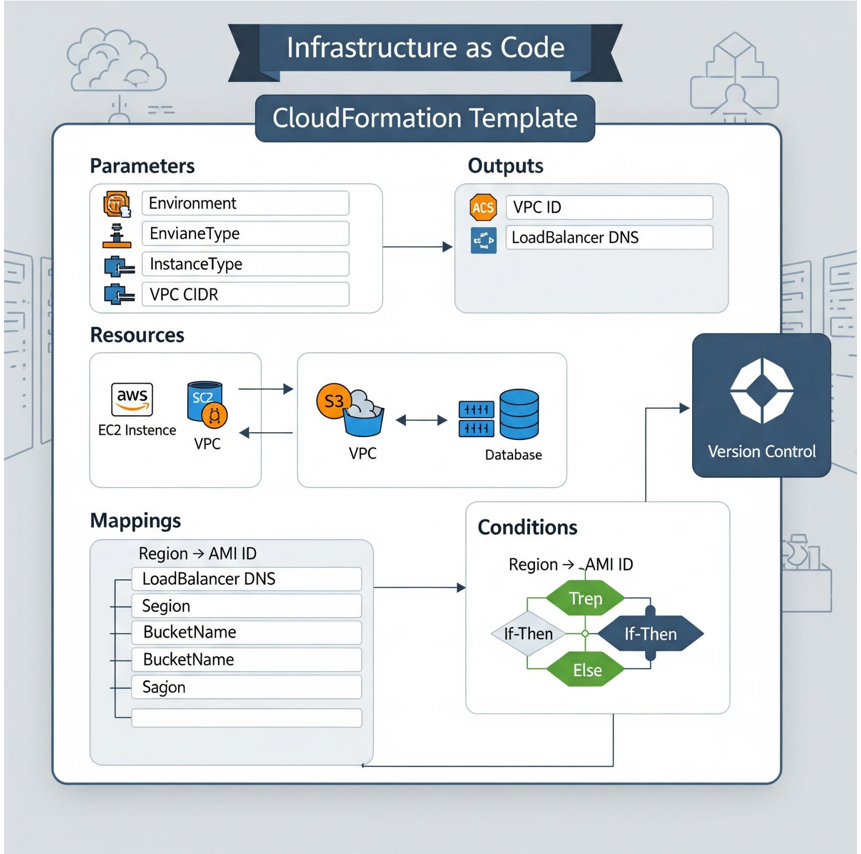
AWS CloudFormation

Cloud Formation Overview

- Infrastructure as Code service
- Define AWS resources in templates
- Supports JSON & YAML
- Automates provisioning at scale
- Enables repeatable deployments

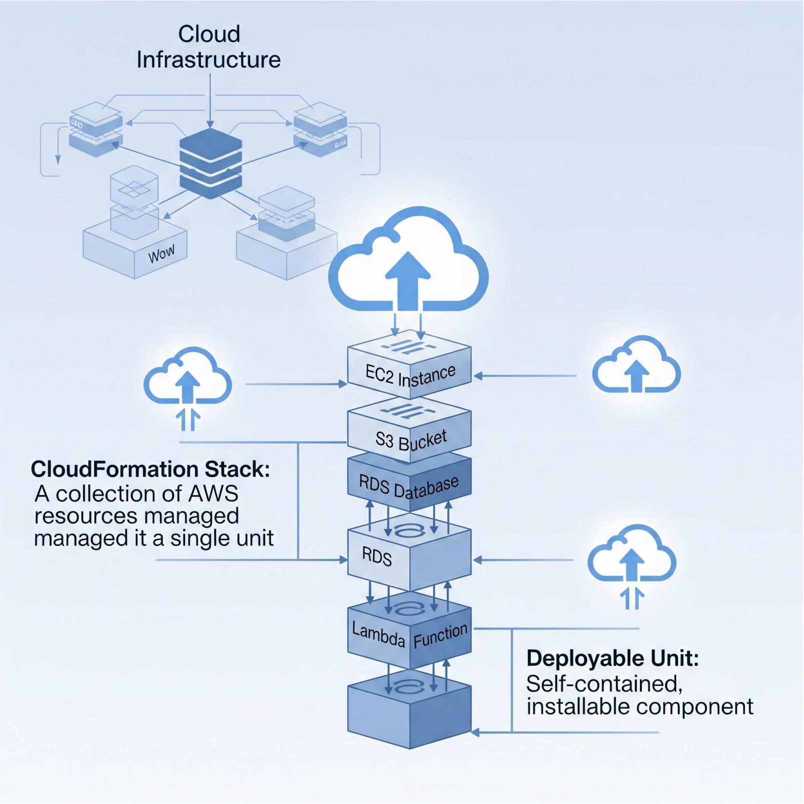
CloudFormation Templates

- Parameters for customization
- Resources define infra components
- Outputs expose values (e.g., VPC ID)
- Mappings & conditions for flexibility
- Stored in version control



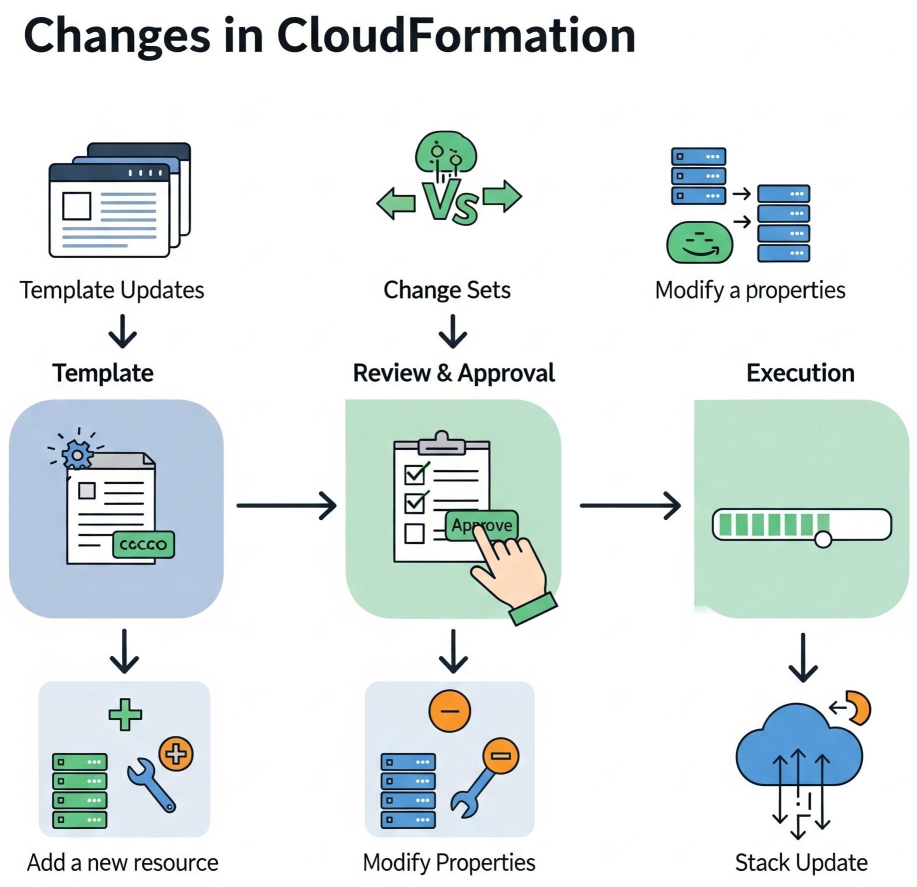
CloudFormation Stacks

- ❖ Stack = deployable unit of resources
- ❖ Create, update, or delete as one
- ❖ Rollback on failure enabled
- ❖ Nested stacks for modularity
- ❖ StackSets for multi-region/multi-account



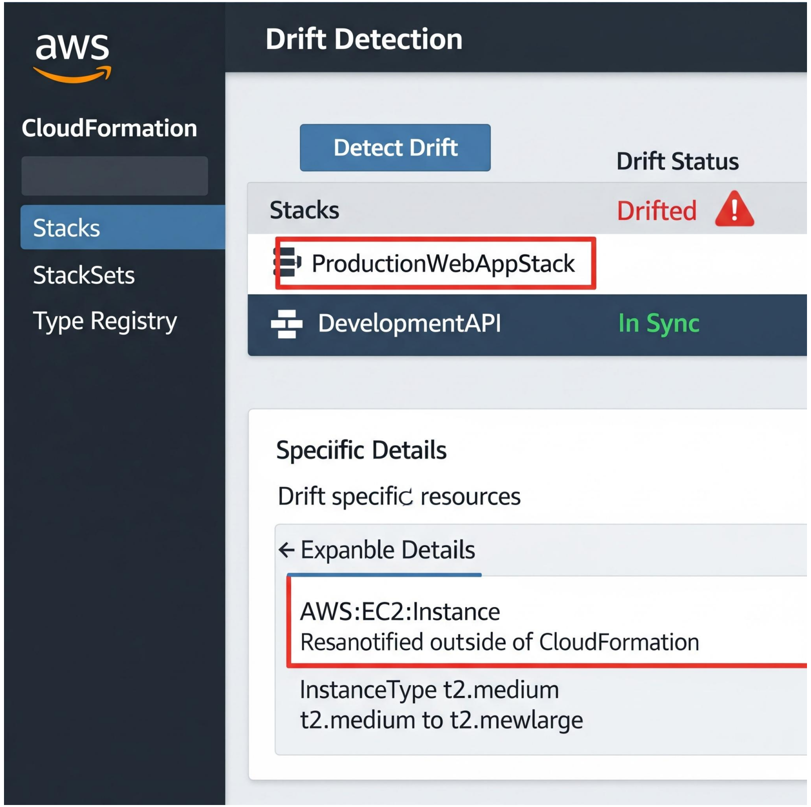
Change Sets

- Preview changes before applyin
- Shows add/modify/delete action
- Helps avoid downtime & errors
- Supports approval workflows
- Best practice before stack updat



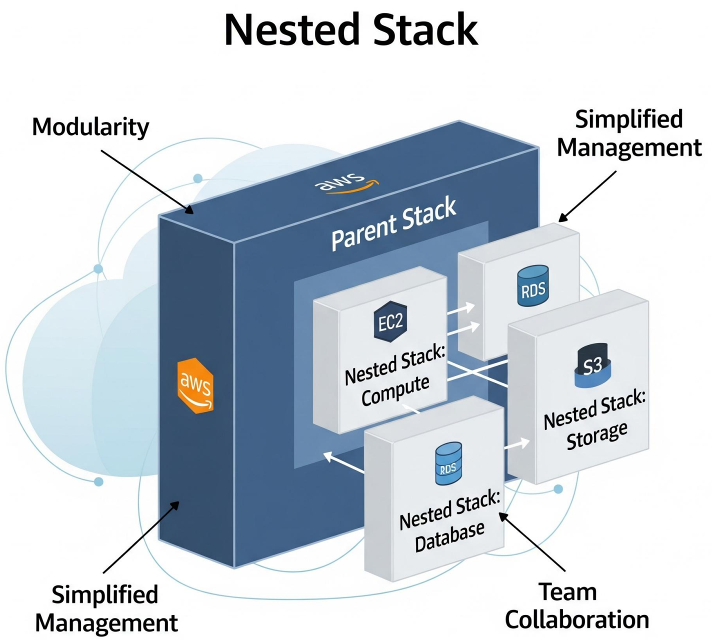
Drift Detection

- ❖ Detects mismatches between template & reality
- ❖ Identifies manual changes in console
- ❖ Ensures compliance with IaC
- ❖ Reduces configuration drift
- ❖ Can remediate via stack update



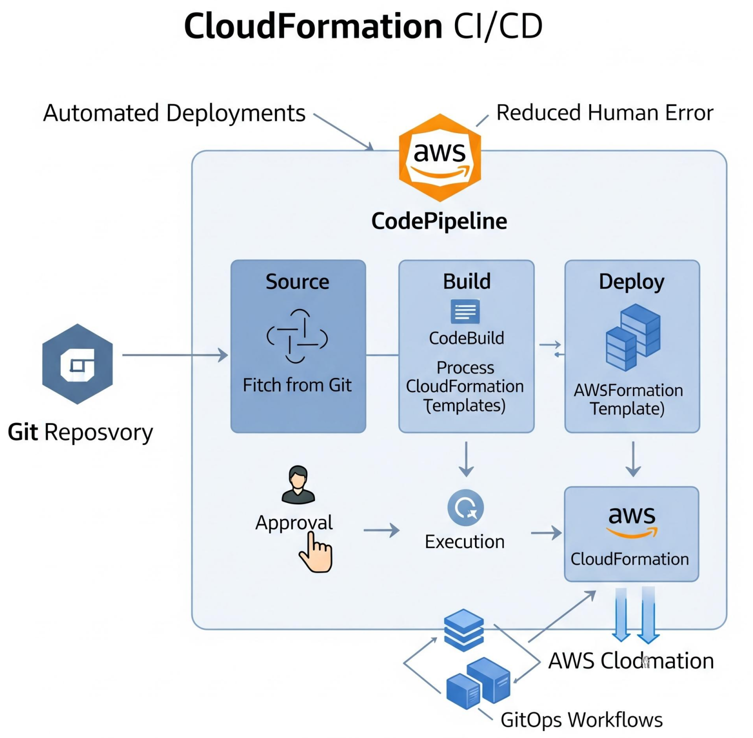
Nested Stacks

- ❖ Break down large templates
- ❖ Promote reusability across teams
- ❖ Easier to maintain & debug
- ❖ Modular approach to IaC
- ❖ Recommended for enterprise-scale infra



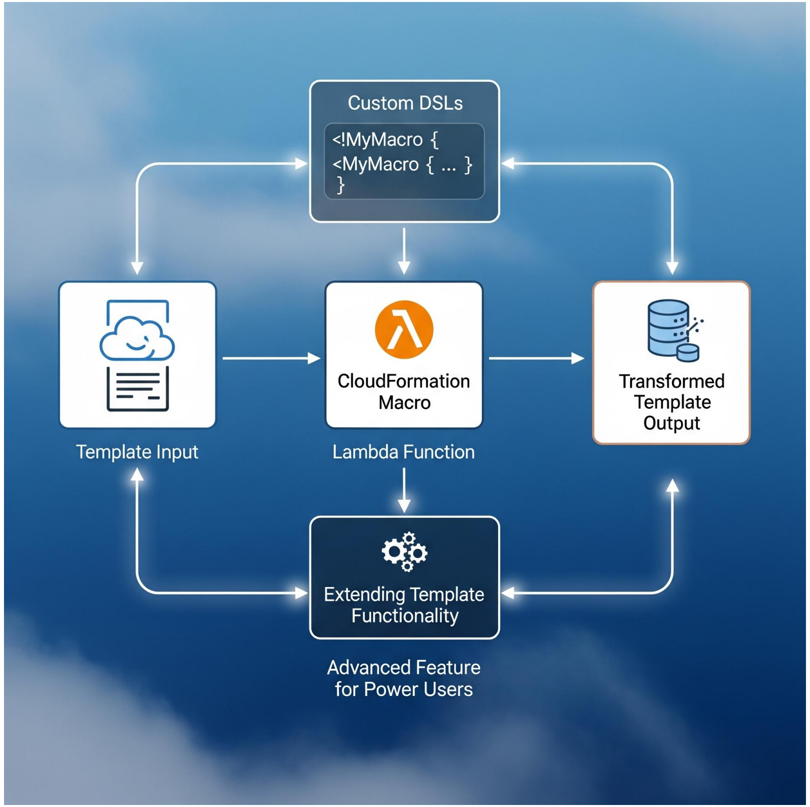
CloudFormation & CI/CD

- Integrates with CodePipeline & CodeBuild
- Automates deployments on commit
- Enables GitOps workflows
- Supports approval stages
- Reduces human error



CloudFormation Macros

- Extend template functionality
- Backed by Lambda functions
- Transform templates dynamically
- Support custom DSLs
- Advanced feature for power users



Best Practices

- ❖ Always use version control
- ❖ Validate templates with linters
- ❖ Use parameters & mappings wisely
- ❖ Apply least privilege IAM roles
- ❖ Test in non-prod first

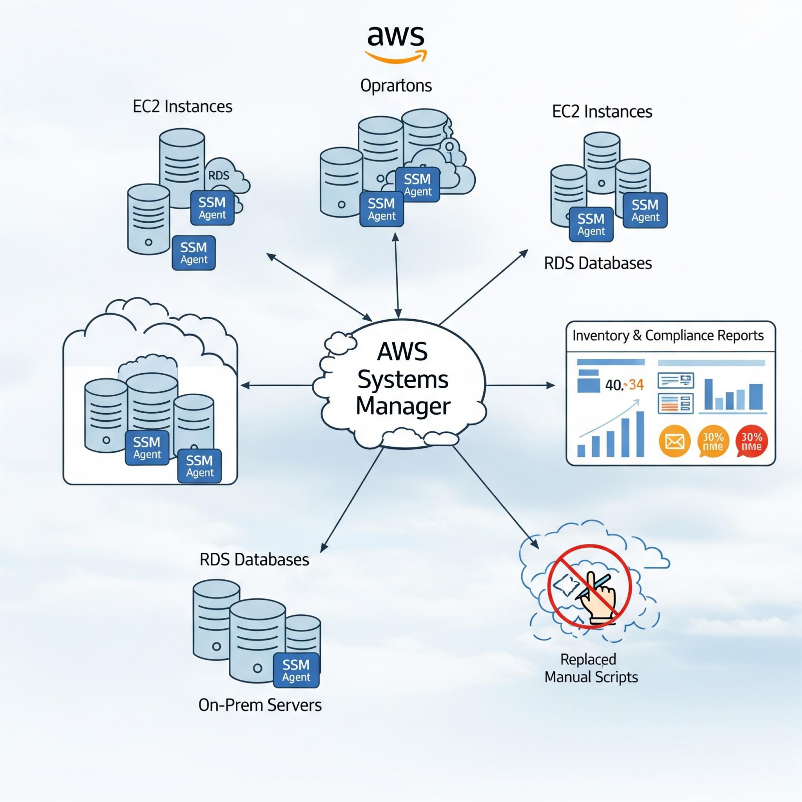
CloudFormation Summary

- ❖ IaC ensures reliability & repeatability
- ❖ Templates define entire infra stack
- ❖ Supports CI/CD integration
- ❖ Reduces drift & manual errors
- ❖ Scales infra management

AWS Systems Manager

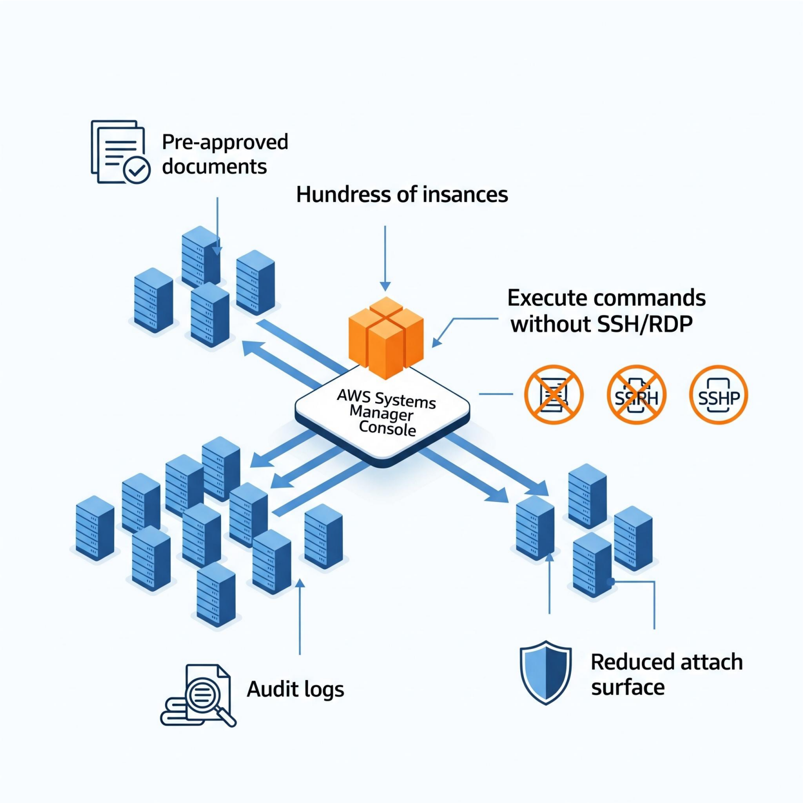
Systems Manager Overview

- Central operations hub for AWS resources
- Manage EC2, RDS, and on-prem servers
- Provides inventory & compliance reports
- SSM Agent runs on instances
- Replaces manual ops scripts



Run Command

- ❖ Execute commands without SSH/RDP
- ❖ Works across hundreds of instances
- ❖ Supports pre-approved documents
- ❖ Provides audit logs
- ❖ Reduces attack surface

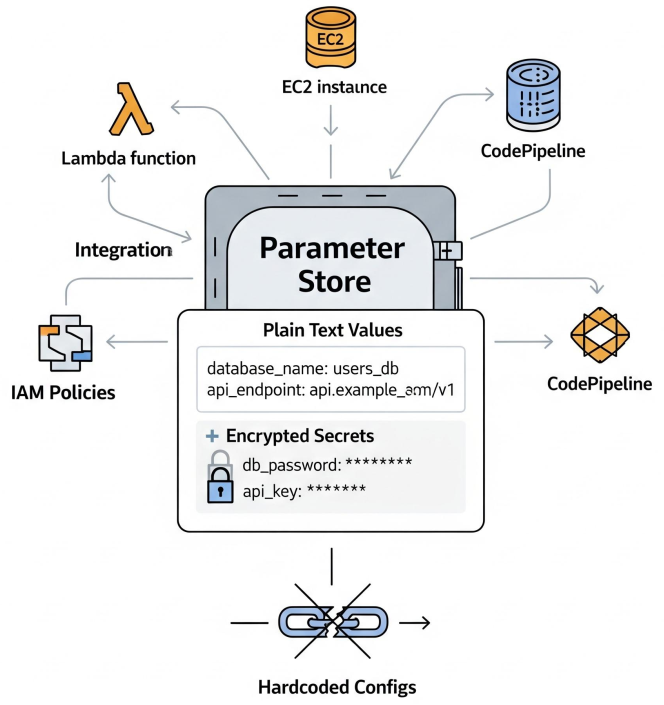


Patch Manager

- Automates OS & software updates
- Schedule patching windows
- Enforce compliance baselines
- Integrates with approval workflows
- Reduces security vulnerabilities

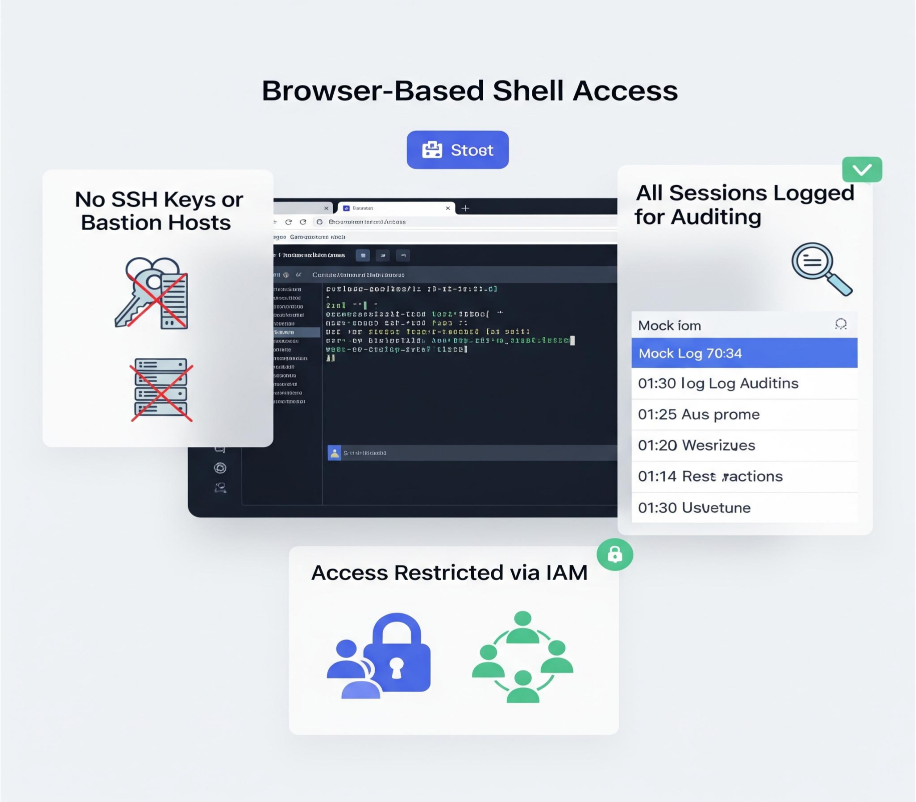
Parameter Store

- ❖ Centralized config & secret storage
- ❖ Supports plain text & encrypted values
- ❖ Integrated with IAM policies
- ❖ Works with Lambda, EC2, CodePipeline
- ❖ Replaces hardcoded configs



Session Manager

- Secure browser-based shell access
- No need for SSH keys or bastion hosts
- Logs all sessions for auditing
- Restrict access via IAM
- Improves compliance posture



Automation Documents

- ❖ Define reusable operational workflows
- ❖ Pre-built runbooks available
- ❖ Automates patching, snapshots, scaling
- ❖ Integrates with CloudWatch alarms
- ❖ Reduces manual runbooks

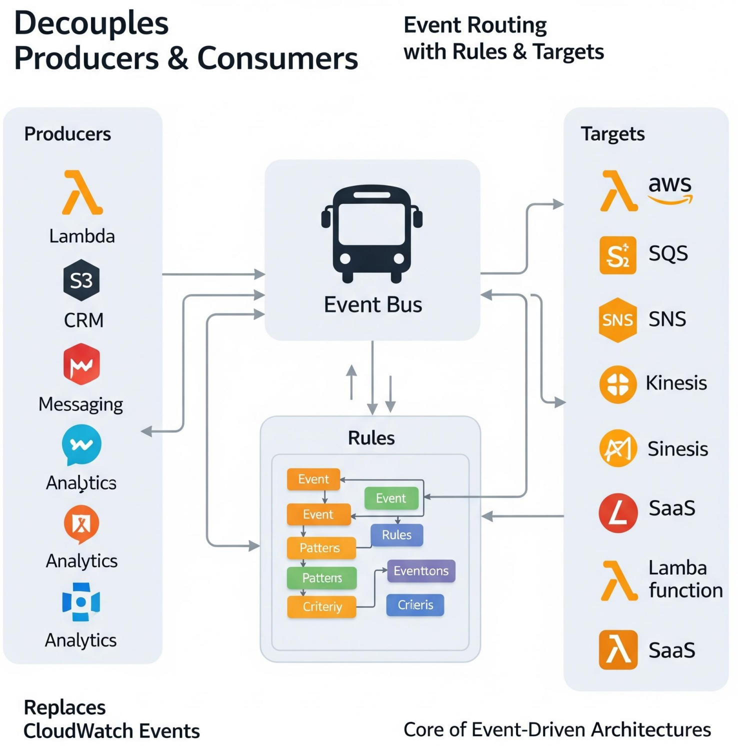
Systems Manager Summary

- ❖ Simplifies ops at scale
- ❖ Secure remote management
- ❖ Centralized secrets & configs
- ❖ Automated patching & workflows
- ❖ Reduces manual admin work

Amazon EventBridge

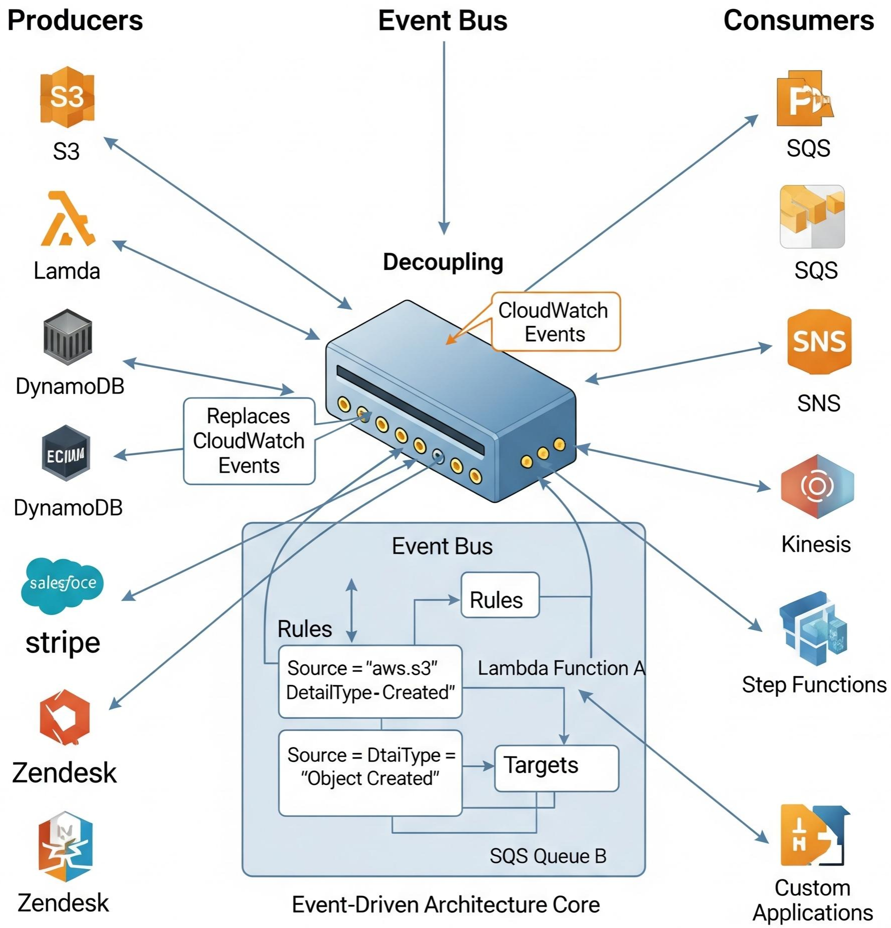
EventBridge Overview

- ❖ Event bus for AWS services & SaaS apps
- ❖ Decouples producers & consumers
- ❖ Event routing with rules & targets
- ❖ Replaces CloudWatch Events
- ❖ Core of event-driven architectures



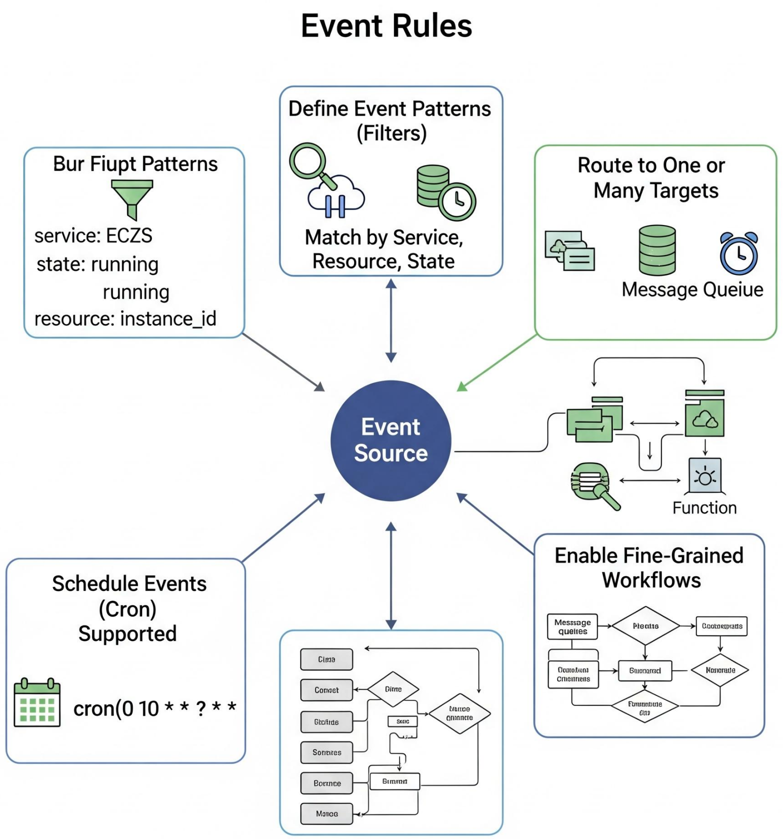
Event Buses

- ❖ Default bus for AWS events
- ❖ Custom buses for apps
- ❖ Partner buses for SaaS integration
- ❖ Events follow JSON schema
- ❖ Enable secure routing



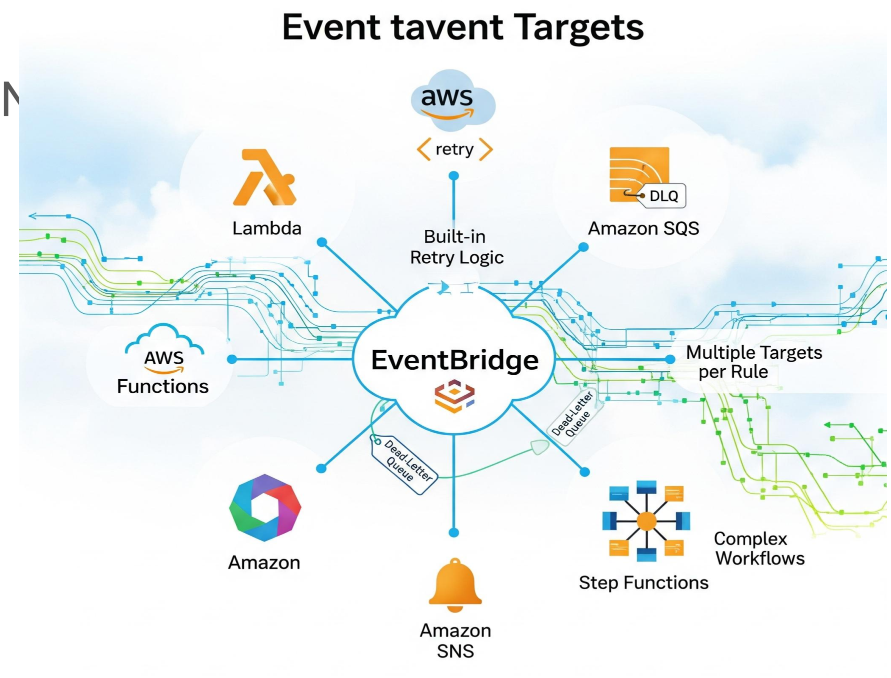
Event Rules

- ❖ Define event patterns (filters)
- ❖ Match by service, resource, state
- ❖ Route to one or many targets
- ❖ Schedule events (cron) supported
- ❖ Enable fine-grained workflows



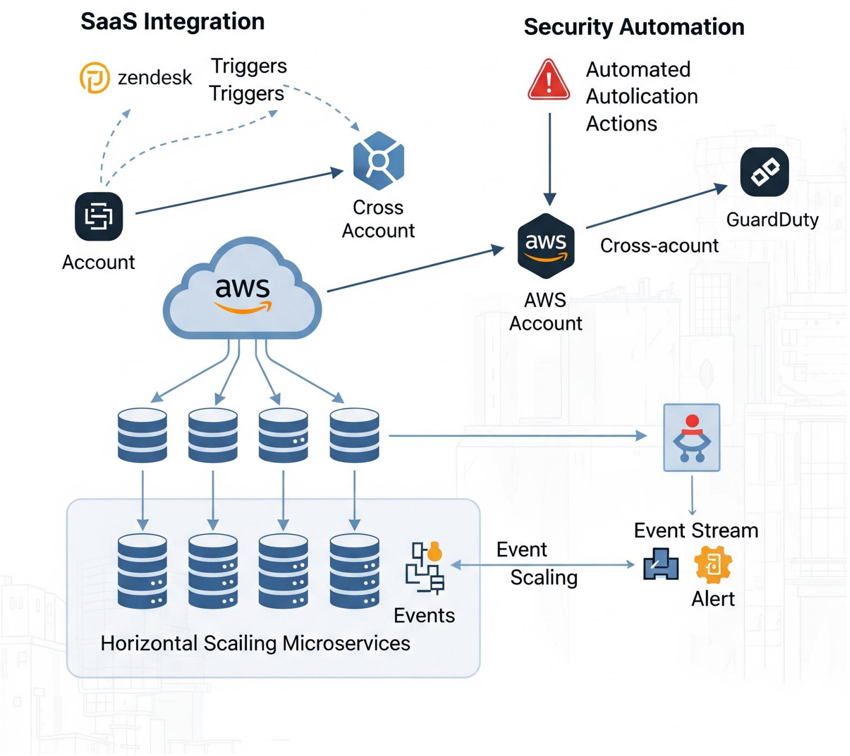
Event Targets

- Targets include Lambda, SQS, SN
  Step Functions
- Retry logic built-in
- Dead-letter queues supported
- Multiple targets per rule
- Enable complex workflows



EventBridge Scenarios

- Auto-scale microservices on events
- SaaS integration triggers (e.g., Zendesk ticket)
- Security automation with GuardDuty alerts
- Connect multiple AWS accounts
- Build loosely coupled apps



EventBridge Summary

- ❖ Flexible, event-driven routing
- ❖ Reduces custom polling scripts
- ❖ Supports SaaS & AWS services
- ❖ Integrates with automation targets
- ❖ Key for modern architectures

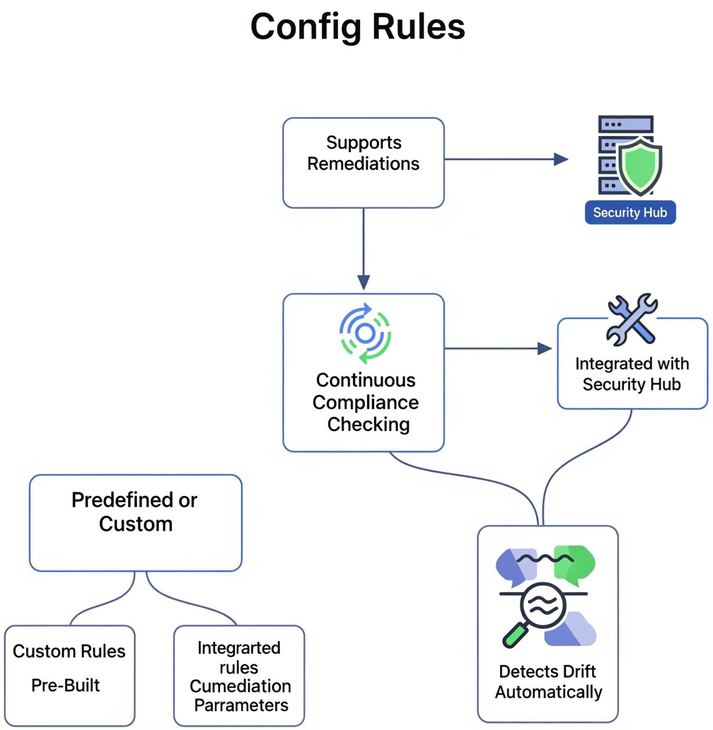
AWS Config

AWS Config Overview

- ❖ Records AWS resource configurations
- ❖ Tracks config changes over time
- ❖ Supports compliance audits
- ❖ Provides managed rules
- ❖ Essential for governance

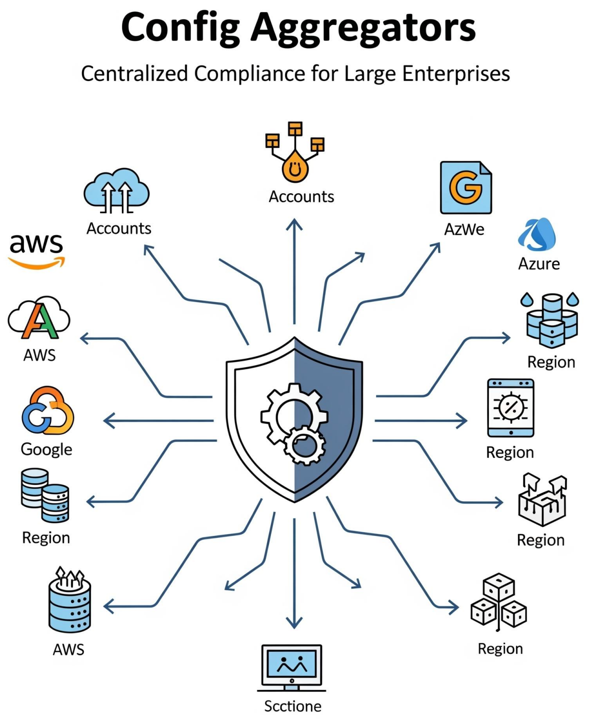
Config Rules

- Predefined or custom rules
- Continuous compliance checking
- Supports remediation actions
- Integrated with Security Hub
- Detects drift automatically



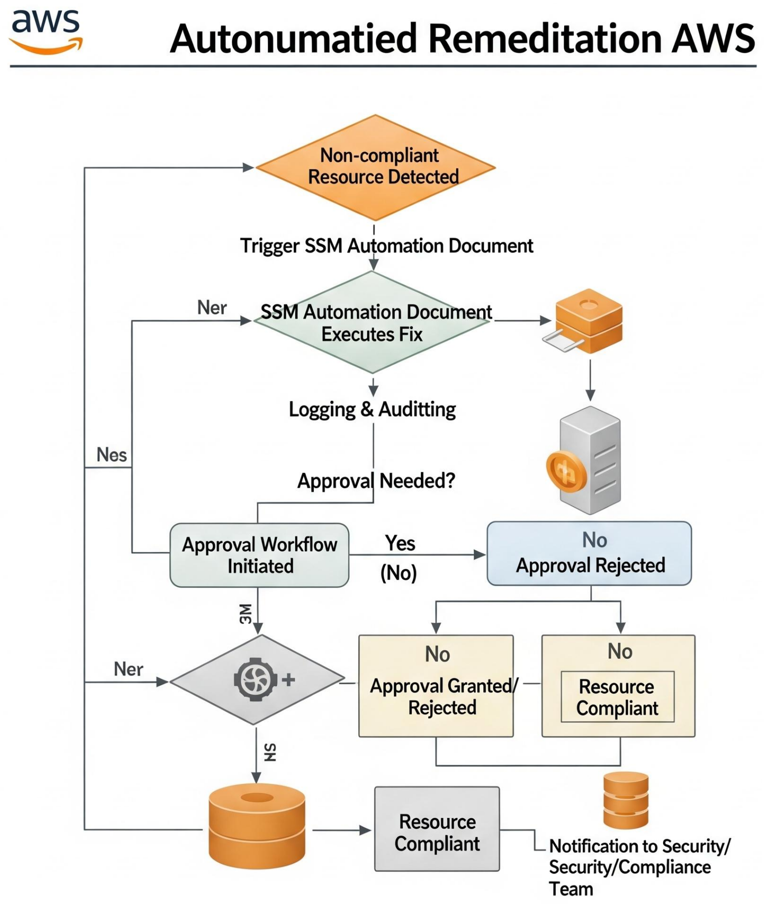
Config Aggregators

- ❖ Collect config across accounts/regions
- ❖ Centralized compliance view
- ❖ Multi-account reporting
- ❖ Simplifies audits
- ❖ Useful for large enterprises



Remediation

- ❖ Automate fixes when non-compliant
- ❖ Uses SSM Automation Docs
- ❖ Prevents manual intervention
- ❖ Improves security posture
- ❖ Supports approvals if needed



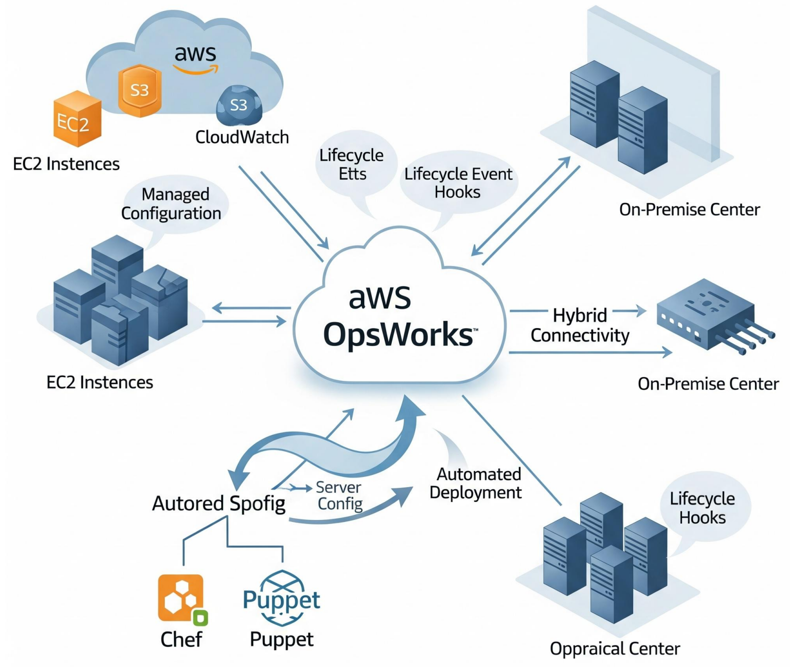
Config Summary

- ❖ Continuous compliance service
- ❖ Tracks, reports, remediates
- ❖ Multi-account governance support
- ❖ Reduces audit effort
- ❖ Key for regulated industries

AWS OpsWorks

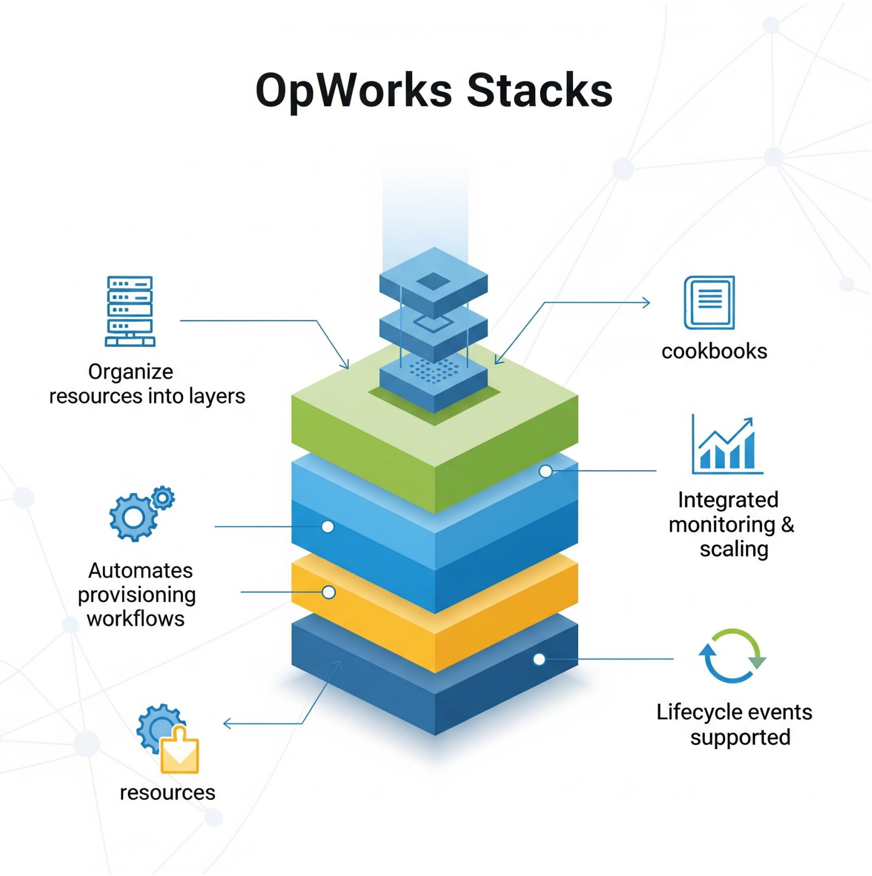
AWS OpsWorks

- ❖ Managed configuration management
- ❖ Supports **Chef & Puppet**
- ❖ Automates server config & deployment
- ❖ Lifecycle event hooks
- ❖ Works with hybrid/on-prem



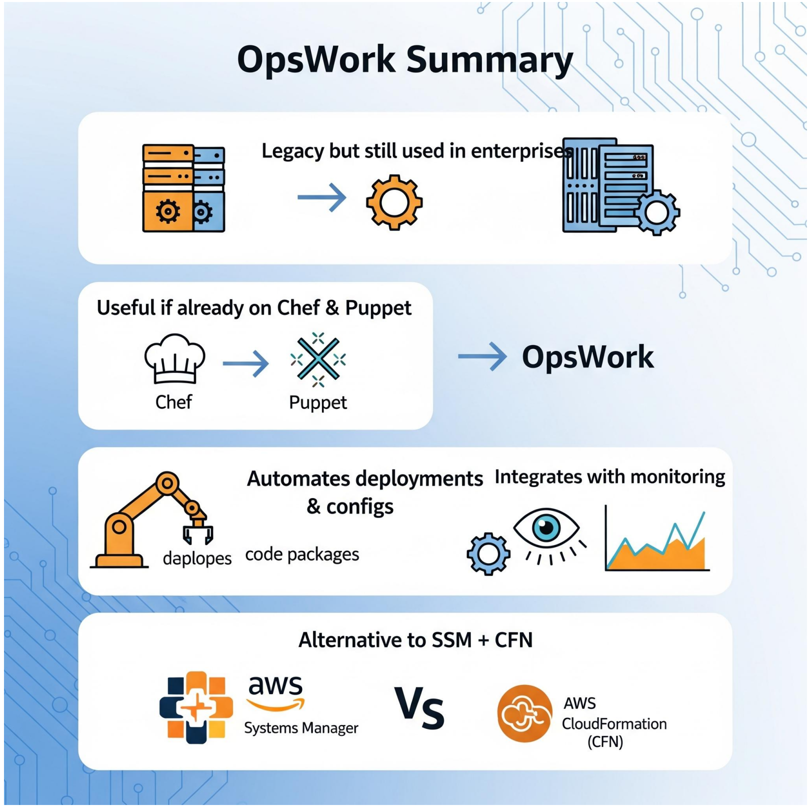
OpsWorks Stacks

- Organize resources into layers
- Automates provisioning workflows
- Custom cookbooks/manifests
- Integrated monitoring & scaling
- Lifecycle events supported



OpsWorks Summary

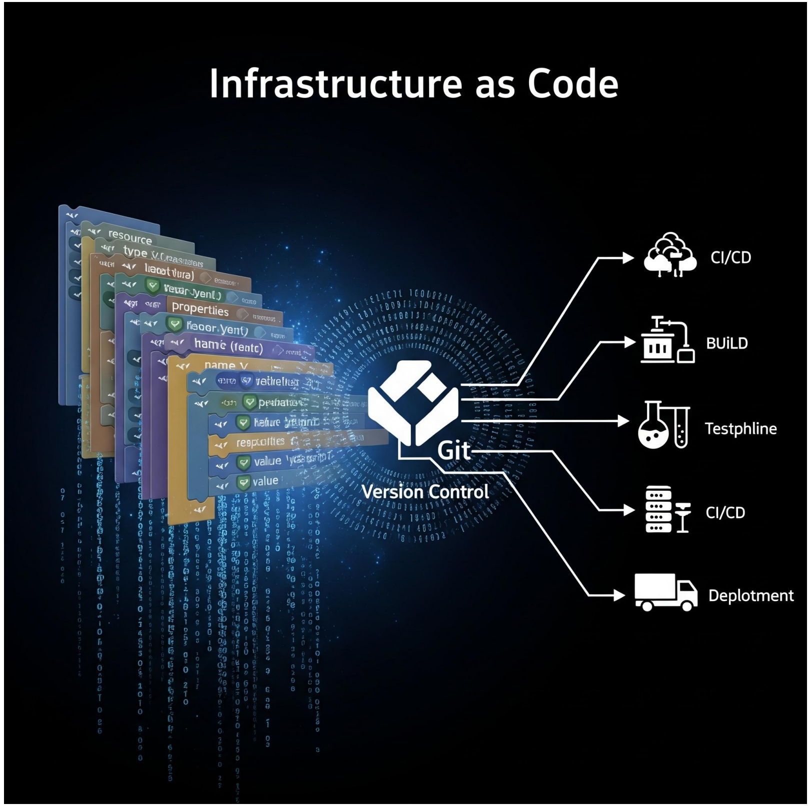
- Legacy but still used in enterprises
- Useful if already on Chef/Puppet
- Automates deployments & configs
- Integrates with monitoring
- Alternative to SSM + CFN



IaC & Automation

Infrastructure as Code (IaC)

- ❖ Define infra in declarative templates
- ❖ Version-controlled infrastructure
- ❖ Enables CI/CD workflows
- ❖ Eliminates manual drift
- ❖ Scales infra reliably

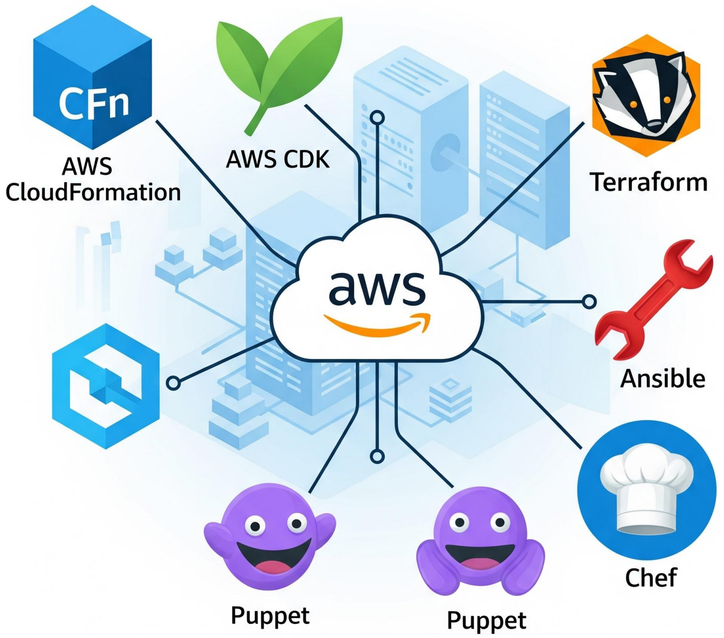


IaC Tools in AWS

- AWS CloudFormation
- AWS CDK (define infra in code)
- Terraform (multi-cloud IaC)
- Ansible/Puppet/Chef integrations
- Each has trade-offs

Automation Scenarios

- ❖ Auto-patch EC2 fleet with SSM
- ❖ Auto-scale EC2 using CloudWatch alarms
- ❖ Enforce compliance with AWS Config
- ❖ Auto-remediate security issues
- ❖ Deploy infra via pipelines



Conclusion

- ❖ AWS provides complete observability stack
- ❖ Automation reduces risk & saves time
- ❖ Use IaC for repeatability
- ❖ Integrate services for end-to-end ops
- ❖ Next: Hands-on labs & real deployments